

## Lecture 23

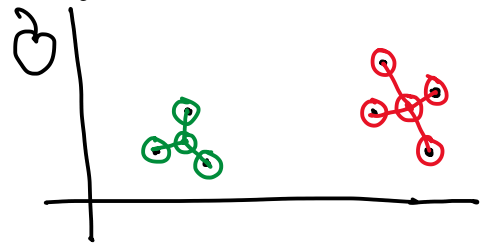
Monday, May 4, 2026 7:21 PM

Recap: K-Means Clustering (unsupervised)

- ① Dataset :  $\{x^{(i)}\}_{i=1}^n$
- ② Input: Integer  $k \in \mathbb{N}$
- ③ A Clustering assigns each datapoint to exactly one cluster  $C: \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, k\}$
- ④  $i \in C_l \Rightarrow$  point  $i$  belongs to cluster  $l$

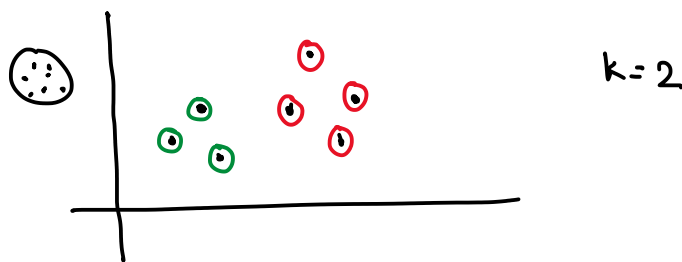
K-Means Objective : minimizing sum of square errors b/w each point & its cluster center

- ①  $\mu_l = \frac{1}{m_l} \sum_{i \in C_l} x^{(i)}$  : centroid of cluster  $l$   
 $\underbrace{m_l}_{\# \text{ of points in } C_l}$
- ②  $SSE = \sum_{i=1}^n \|x^{(i)} - \mu_{c(i)}\|^2$

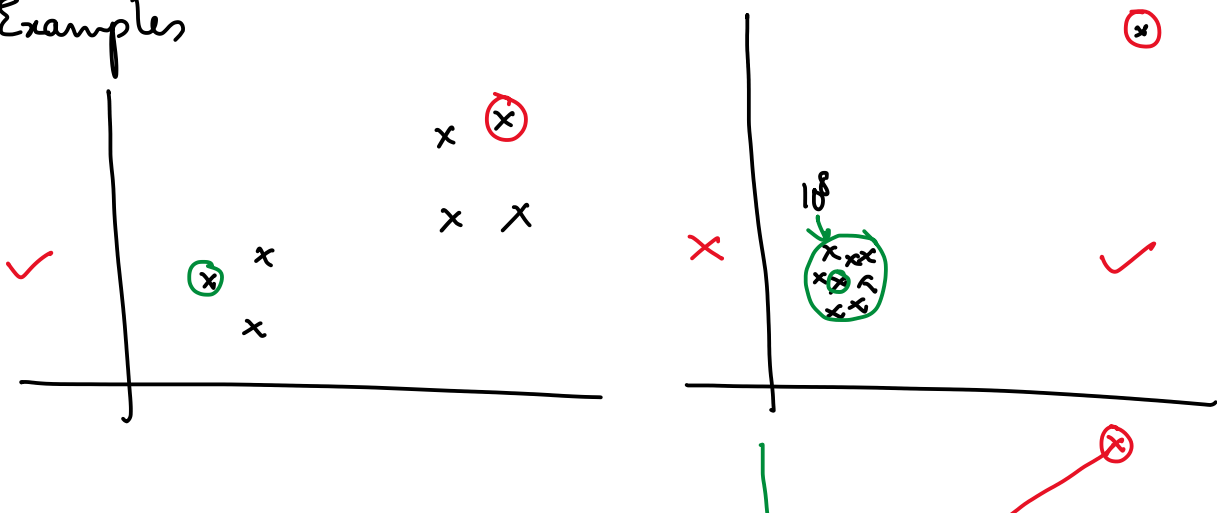


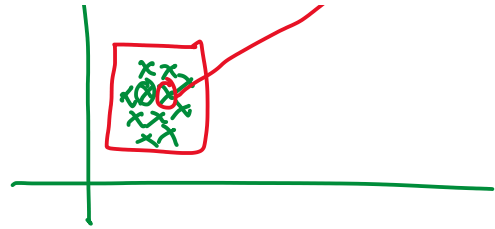
where  $\mu_{c(i)}$  = centroid of the cluster containing point  $i$

- ③ Cluster that minimizes SSE is the best cluster



Examples





Iterative process

- a) Assignment: each point is assigned to the nearest centre
- b) Update: cluster centres are update

Theorem (K-Means): The K-Means clustering algo. terminates in a finite number of steps.

Proof outline: Iteration  $t$   
Iteration  $t+1$

→  $SSE(t)$ :  $SSE$  at the end of iteration  $t$  ←

→  $\underbrace{SSE(t+1, a)}_{II}$ :  $SSE$  value at the end of the assignment step of iteration  $t+1$

$$\begin{aligned} ① \quad \underbrace{SSE(t+1, a)}_{II} &< \underbrace{SSE(t)}_{I} \quad \text{— assignment reduces } SSE \\ ② \quad \underbrace{SSE(t+1)}_{III} &< \underbrace{SSE(t+1, a)}_{II} \quad \text{— update reduces } SSE \end{aligned}$$

$$① \& ② \Rightarrow \underbrace{SSE(t+1)}_{III} < \underbrace{SSE(t)}_{I}$$

The number possible clusterings is finite:  $k^n$

What does it converge to?

— Local minima

Solutions: good initialization  
try multiple initializations, pick the one with the lowest SSE

$$\begin{aligned} C: \{1, 2, 3, 4, \dots, n\} &\rightarrow \{1, 2, \dots, k\} \\ C(1) \in \{1, 2, \dots, k\} \\ C(2) \in \{1, 2, \dots, k\} \end{aligned} \quad \left. \vphantom{\begin{aligned} C(1) \in \{1, 2, \dots, k\} \\ C(2) \in \{1, 2, \dots, k\} \end{aligned}} \right\} = k^n$$

## 6 Choosing the Number of Clusters $k$

The number of clusters  $k$  is a **hyperparameter** that must be chosen by the user. This is one of the most practically important and difficult choices in clustering.

### 6.1 The Elbow Method

A common heuristic is to plot SSE as a function of  $k$  and look for an “elbow” — a point of diminishing returns where increasing  $k$  no longer substantially reduces SSE.

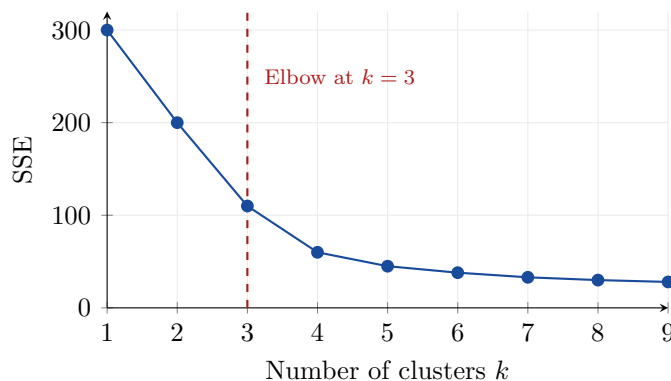


Figure 3: The elbow method. SSE always decreases as  $k$  increases (adding more clusters can only help). The elbow at  $k = 3$  suggests 3 is a natural number of clusters for this dataset.

#### Limitations of the Elbow Method

The elbow is often not sharp — the plot may be smooth with no obvious kink. In such cases, domain knowledge or other criteria (e.g. silhouette score, gap statistic) are needed to choose  $k$ .

### 6.2 Domain Knowledge

In many applications, the number of clusters has a natural meaning: a retailer may want exactly 5 customer segments for operational reasons; a biologist may know there are 3 cell types. When such context is available, it should take precedence over data-driven heuristics.

⋮

How do fix  $k$ ?

Use the elbow method.

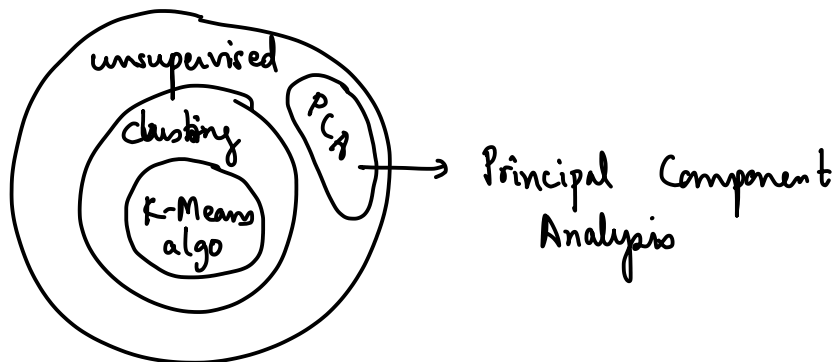
Classification

$$f: \begin{matrix} x \rightarrow y \\ \downarrow \quad \downarrow \\ \mathbb{R}^d \quad \{0, 1, \dots, k-1\} \end{matrix}$$

using  $\{x^{(i)}, y^{(i)}\}_{i=1}^n$

Clustering:

$$C: \{1, 2, \dots, n\} \rightarrow \{0, 1, \dots, k-1\}$$



Linear Algebra Refresher

- Matrix  $\begin{matrix} \xrightarrow{\text{rows}} \\ \xrightarrow{\text{columns}} \end{matrix}$

$n$  rows  
 $m$  columns

$$\begin{bmatrix} a_{ij} \end{bmatrix}$$

$i^{\text{th}}$  row  
 $j^{\text{th}}$  column

$n=3 \quad m=2$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

- Square matrix:  $n=m$

- Transpose of a matrix

$A: a_{ij} \quad A^T: a_{ij}^T = a_{ji}$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$



- Symmetric matrix:  $A = A^T$   
 $n=m$

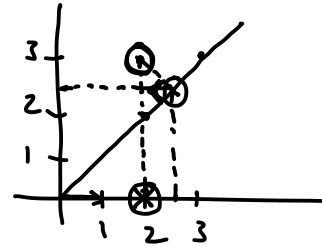
## - Projections

$$x \in \mathbb{R}^d$$

$\rightarrow u \in \mathbb{R}^d$  - unit vector  $\|u\|_2 = 1$   $u^T u = 1$

$$\text{proj}_u(x) = (u^T x) u \quad \text{--- (1)}$$

$$d=2, x = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$\begin{aligned} \text{proj}_{u_1}(x) &= \left( \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= (1 \times 2 + 0 \times 3) \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \end{aligned}$$

$$u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\|u_2\|_2 = 1$$

$$\begin{aligned} \text{proj}_{u_2}(x) &= \left( \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right) \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \\ &= \frac{5}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 5/2 \\ 5/2 \end{bmatrix} \end{aligned}$$

## - Eigenvalues & Eigenvectors

Given

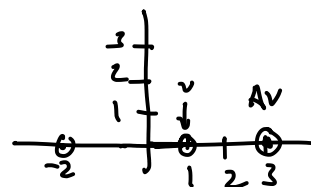
$$A \in \mathbb{R}^{m \times m}, \quad v \neq 0 \quad v \in \mathbb{R}^m, \quad \lambda \in \mathbb{R}$$

if  $A v = \lambda v$ , then  $\lambda$  is an eigenvalue of  $A$   
 $v$  is the corresponding eigenvector of  $A$

Diagram showing matrix dimensions:  $A$  is  $m \times m$ ,  $v$  is  $m \times 1$ , and  $\lambda v$  is  $m \times 1$ .

Example:  $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \quad v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$A v = \begin{pmatrix} 3 \\ 0 \end{pmatrix} = 3 v$$



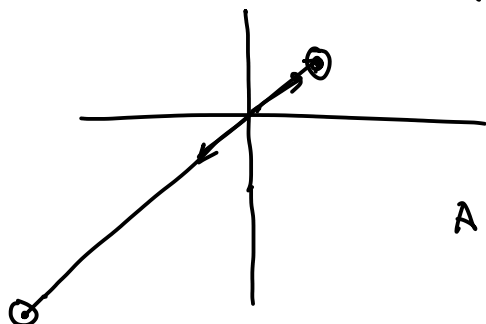
$$v = \begin{pmatrix} 1 \\ 0 \end{pmatrix} : \text{eigenvector of } A \quad \lambda = 3$$

$$\underbrace{A v}_{\substack{\uparrow \\ \text{eigenvector}}} = \underbrace{\lambda v}_{\text{scaled version of the original vector}}$$

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \quad v = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad Av = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \lambda = 2$$

$$Av = \lambda v$$

$A$   $m \times m$   
 $v$   $m \times 1$  could be  
 $\lambda \in \mathbb{R}$  (-ve)



$$\lambda = -5$$

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$$

How to compute eigenvalues & eigenvectors?

①  $Av = \underline{\lambda v}$  RHS <sup>(a)</sup>  $\lambda \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \end{pmatrix}$

$= \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \end{pmatrix}$

②  $Av = \lambda I v$  where  $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  (b)

$$Av - \lambda I v = 0$$

$$\Rightarrow \underbrace{(A - \lambda I)}_{m \times m} \underbrace{v}_{m \times 1} = 0 \quad v \neq 0$$

For non-zero  $v$ , matrix  $A - \lambda I$  must be singular  
 $\Rightarrow \det(A - \lambda I)$  should be zero.

Defn: (postponed to next class)

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \det(A) = a_{11}a_{22} - a_{21}a_{12}$$

$$\det(A - \lambda I) = 0 \quad A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix}$$

$$\Rightarrow (a_{11} - \lambda)(a_{22} - \lambda) - a_{21}a_{12} = 0 \quad \leftarrow \text{equation to be solved}$$

↳ Solve this equation to get  $\lambda$